

internal resistance therefore does not affect the measurement of e.m.f. of X.

22 Ans: A

$$\text{Potential at X: } V = \frac{-2Q}{4\pi\epsilon_0 r} + \frac{Q}{4\pi\epsilon_0 r} = -\frac{Q}{4\pi\epsilon_0 r}$$

$$\text{Magnitude of E-field at X: E due to } -2Q \text{ charge} = \frac{2Q}{4\pi\epsilon_0 r^2}$$

$$\text{E due to } +Q \text{ charge} = \frac{Q}{4\pi\epsilon_0 r^2}$$

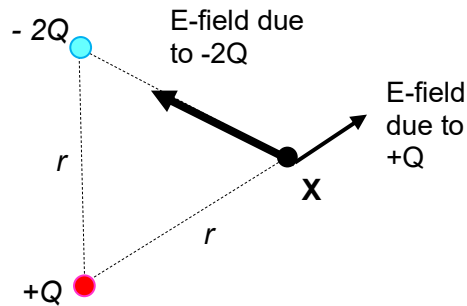
If these 2 vectors were in a straight line and “head to tail” ,

$$\text{then the largest possible magnitude of the total E-field} = \frac{2Q}{4\pi\epsilon_0 r^2} + \frac{Q}{4\pi\epsilon_0 r^2} = \frac{3Q}{4\pi\epsilon_0 r^2}$$

If these 2 vectors were in a straight line and “head to head” ,

$$\text{then the minimum possible magnitude of the total E-field} = \frac{2Q}{4\pi\epsilon_0 r^2} - \frac{Q}{4\pi\epsilon_0 r^2} = \frac{Q}{4\pi\epsilon_0 r^2}$$

Direction of E: At point X,



The resultant of these 2 E-fields would point upwards and to the left

23 Ans: B